

Topic 4

Coordinate geometry

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions**

Learning focus

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

4.1	Coordinates and gradients
4.2	Distance and midpoint
4.3	Equations of straight lines
4.4	Parallel and perpendicular lines
4.5	Intersection and geometric reasoning
4.6	Graphical intersection checks

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Gradient from two points

The gradient of the line from (x_1, y_1) to (x_2, y_2) is change in y divided by change in x . A positive gradient rises to the right and a negative gradient falls. A horizontal line has gradient zero; a vertical line has undefined gradient.

Worked example

From $(2, 3)$ to $(6, 11)$, $m = (11 - 3)/(6 - 2) = 2$.

Exam check: Keep coordinate subtraction in the same order in numerator and denominator.

2. Midpoint and distance

The midpoint averages the two x -values and the two y -values separately. The distance formula comes from Pythagoras: $d^2 = (\Delta x)^2 + (\Delta y)^2$. A sketch helps detect an impossible negative distance or midpoint outside the segment.

Worked example

From $(1, 2)$ to $(7, 10)$, midpoint = $(4, 6)$ and distance = $\sqrt{6^2 + 8^2} = 10$.

Exam check: Distance is non-negative, even if the line has a negative gradient.

3. Equation of a line

Use $y = mx + c$ when the gradient and vertical intercept are known. If a point (x_1, y_1) and m are given, substitute into $y - y_1 = m(x - x_1)$, then simplify. To find x -intercept set $y = 0$; to find y -intercept set $x = 0$.

Worked example

Gradient 3 through $(2, 7)$: $y - 7 = 3(x - 2)$, hence $y = 3x + 1$.

Exam check: Substitute the original point to check the final equation.

4. Parallel and perpendicular lines

Parallel non-vertical lines have equal gradients. Perpendicular non-vertical lines have gradients whose product is -1 . Horizontal and vertical lines are perpendicular to each other, even though the vertical gradient is undefined.

Worked example

A line perpendicular to $y = 2x + 1$ has gradient $-1/2$. Through $(0, 3)$ its equation is $y = -x/2 + 3$.

Exam check: Taking the negative alone is not enough; take the negative reciprocal.

5. Intersection and coordinate proofs

The intersection of two lines satisfies both equations. Solve simultaneously, then substitute the result into both. To prove a quadrilateral is a parallelogram, compare gradients of opposite sides or find matching midpoints of its diagonals. To check right angles, multiply gradients.

Worked example

For $y = 2x + 1$ and $y = -x + 7$, $2x + 1 = -x + 7$ gives $x = 2$; $y = 5$.

Exam check: An accurately drawn graph supports a result, but use algebra if exact coordinates are requested.

6. Graphical intersection checks

A graph can suggest where two roads or lines intersect; solve the corresponding equations to obtain an exact coordinate. Sketch or plotting tools help choose a suitable region, but a pair that misses either original equation is incorrect.

Worked example

Lines $y = x + 1$ and $y = -2x + 7$ intersect where $x + 1 = -2x + 7$: $(2, 3)$.

Exam check: Check both original equations after a graphical estimate.

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Graphical calculator and Paper 6

Plot points with equal axis scales when inspecting shape. Use slope and distance formulae to support a graphical observation.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.

Practice paper

Topic 4 · Coordinate geometry

Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:	
Class / date:	

Total: 39 marks

Section A · Multiple choice

Choose one answer for each question.

1. Gradient from (0,1) to (2,5)? [1]

- A 1
- B 2
- C 3
- D 4

2. Midpoint of (2,4) and (8,10)? [1]

- A (6,14)
- B (5,7)
- C (3,3)
- D (10,14)

3. Distance between (0,0) and (3,4)? [1]

- A 4
- B 5
- C 6
- D 7

4. Which line is parallel to $y = 3x + 2$? [1]

- A $y = -3x + 4$
- B $y = x/3 + 4$
- C $y = 3x - 7$
- D $y = -x/3 + 1$

5. Which gradient is perpendicular to -4? [1]

- A 4
- B $-1/4$
- C $1/4$
- D -4

6. What is the y-intercept of $y = -2x + 9$? [1]

- A -2
- B 2
- C -9
- D 9

7. For $y = 2x - 6$, the x-intercept is [1]

- A -3
- B 0
- C 3
- D 6

8. Which is the equation of the horizontal line through (4,-2)? [1]

- A $x = 4$
- B $y = 4$
- C $x = -2$
- D $y = -2$

Section B · Structured questions

9. A(1,2) and B(7,10) lie on a straight line.

(a) Calculate the gradient of AB.

[2]

(b) Calculate the midpoint of AB.

[2]

(c) Calculate the length AB.

[2]

[Total: 6]

10. A line passes through (2,5) and has gradient -3.

(a) Find its equation in the form $y = mx + c$.

[3]

(b) State its y-intercept.

[1]

(c) Find its x-intercept.

[2]

[Total: 6]

11. The lines $y = 2x + 3$ and $y = -x + 12$ intersect.

(a) Find the x-coordinate of their intersection.

[2]

(b) Find the y-coordinate.

[1]

(c) Check the point satisfies both equations.

[2]

[Total: 5]

12. P(0,4), Q(6,7) and R(8,3) are three vertices of a quadrilateral.

(a) Find the gradient of PQ.

[2]

(b) Find the gradient of QR.

[2]

(c) Explain whether PQ is perpendicular to QR.

[2]

[Total: 6]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. Points A(0,0), B(4,3), C(8,6) form the start of a linear path.

(a) Calculate the gradient of AB and BC. [2]

(b) Suggest coordinates for the next point D if the pattern continues. [2]

(c) Find the distance AD. [2]

(d) State an assumption needed to extend the path further. [2]

[Total: 8]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. B — 2. Change in y is 4 over change in x of 2.
2. B — (5,7). Average corresponding coordinates: $(2+8)/2$ and $(4+10)/2$.
3. B — 5. Pythagoras gives $\sqrt{3^2+4^2} = 5$.
4. C — $y = 3x - 7$. Parallel lines have the same gradient of 3.
5. C — $1/4$. A perpendicular gradient satisfies $(-4)m = -1$.
6. D — 9. Set $x = 0$; the remaining y -value is 9.
7. C — 3. Set $y = 0$ and solve $2x = 6$.
8. D — $y = -2$. A horizontal line has a constant y -coordinate of -2.

Section B • Worked solutions

9. a) $(10 - 2)/(7 - 1) = 4/3$ [2]. b) $((1 + 7)/2, (2 + 10)/2) = (4, 6)$ [2]. c) $\sqrt{6^2 + 8^2} = 10$ [2].

10. a) $y - 5 = -3(x - 2)$, so $y = -3x + 11$ [3]. b) 11 [1]. c) $0 = -3x + 11$, $x = 11/3$ [2].

11. a) $2x + 3 = -x + 12$, $3x = 9$, $x = 3$ [2]. b) $y = 9$ [1]. c) $2(3) + 3 = 9$ and $-3 + 12 = 9$ [2].

12. a) $(7 - 4)/(6 - 0) = 1/2$ [2]. b) $(3 - 7)/(8 - 6) = -2$ [2]. c) $(1/2)(-2) = -1$, therefore perpendicular [2].

Section C • Investigation and modelling solutions

13(a). Both gradients are $3/4$ [2].

13(b). Add (4,3) to C: D(12,9) [2].

13(c). $AD = \sqrt{(12^2 + 9^2)} = 15$ [2].

13(d). The step vector must stay (4,3) or the path remain linear [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.